DECOR: Dynamic Decoupling and Multiobjective Optimization for Long-Tailed Remote Sensing Image Classification

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***Abstract*— In the realm of remote sensing, targets of interest span a range of categories. However, their distribution is not always uniform. Certain categories substantially outnumber others, resulting in what's termed a "long-tailed distribution" in remote sensing imagery. This imbalanced distribution often biases a classifier's focus toward the more abundant (head) classes, at the detriment of the less-represented (tail) classes. Such biases undermine the classifier's generalization performance, particularly in the context of remote sensing image classification (RSIC). While existing mitigation approaches such as resampling, reweighting, and transfer learning offer some respite, they often miss out on in-depth knowledge refinement, rendering them less effective for severe long-tailed RSIC scenarios. To counter these challenges, we introduce DECOR, a dynamic decoupling (DD) and multiobjective optimization framework (MOOF). Within DECOR, the feature extractor and classifier are dynamically decoupled, promoting superior feature representation and classifier training. Then, a multiobjective optimization approach is proposed to delve deeper, refining feature representation at the knowledge level using learnable feature centroids (LFCs) coupled with masked world knowledge learning (WKL). Moreover, to combat the pronounced effects of sample imbalance on classifier training, we employ a class-balanced resampling technique paired with a parameter-efficient adapter, which sharpens the classifier's decision boundary and bridges the gap between representation and classification. DECOR's efficacy is validated through comprehensive experiments on several datasets, including the NWPU-RESISC45-LT (NWPU-LT), AID-LT, and our self-built BIT-AFGR50-LT. Experimental results demonstrate DECOR's marked enhancement in performance on long-tailed datasets. Our source code is available at: https://github.com/ChloeeGrace/DECOR.**

***Index Terms*— Decouple learning, long tail, remote sensing scene classification.**

Manuscript received 30 November 2023; revised 7 January 2024; accepted 31 January 2024. Date of publication 23 February 2024; date of current version 6 March 2024. This work was supported in part by the General Program of the National Natural Science Foundation of China under Grant 62371048, in part by the National Science Foundation for Young Scientists of China under Grant 62101046, and in part by the Multisource Satellite Data Hardware Acceleration Computing Method With Low Energy Consumption under Grant 2021YFA0715204. *(Corresponding author: Guanqun Wang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3369178

I. Introduction

REMOTE sensing image classification (RSIC) is a basic but important task in remote sensing intelligent interpretation. It plays an important role in many fields such as urban planning [1], [2], [3], environmental protection [4], [5], [6], resource exploration [7], and military reconnaissance [8], whose performance directly affects the effectiveness of remote sensing intelligent interpretation systems.

However, in the real world, the sample size of targets in different categories usually distributes unevenly, whose distribution can be commonly demonstrated as the long-tailed curve in Fig. 1, and such data distribution is referred to as "long-tailed distribution." In remote sensing images, the targets of interest are usually characterized by intense long-tailed distributions in the real world (e.g., vehicles, airports plus a statistical map of remote sensing data) and thus the problem of long-tailed image classification is an inescapable research element in remote sensing intelligent interpretation applications.

As shown in Fig. 1, the sample size of different categories in the popular publicly available remote sensing dataset, dataset for object detection in aerial images (DOTA) [9], is counted. It can be clearly seen that the number of targets in different categories is characterized by a long-tailed distribution. According to the distribution of sample size, different categories are divided into three parts: head class, middle class, and tail class. In the training phase, the data imbalance between the head class samples and the tail class samples makes the feature representation of tail class targets difficult to fully explore, which may result in a serious decline in the model generalization ability. When dealing with remote sensing data with long-tailed distribution, since certain categories have a much higher sample size than others, it will lead to uneven feature learning in the network and inevitably tend to learn the target features of the head class with a larger sample size while ignoring the tail class with a smaller one. On the other hand, in the long-tailed classification task, when the number of tail class samples is too small, the classification of tail class samples will be transformed into a few-shot learning problem, which poses a great challenge to the network structure and training strategy.

For a more intuitive explanation, a "difficulty coefficient" D is defined to quantify the challenges in long-tailed

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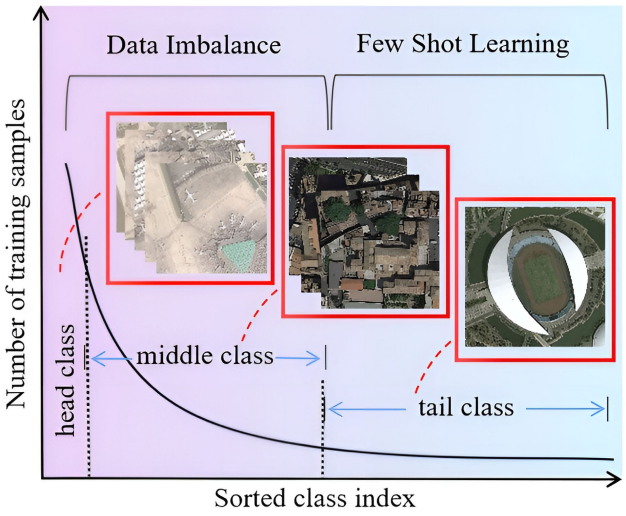


Fig. 1. Long-tailed distribution. At the number of samples level, it usually is classified as head classes, middle classes, and tail classes. At the remote sensing scene classification task level, it can be classified as data imbalance and few-shot learning.

classification, encompassing two key factors: class imbalance and classification difficulty. For any long-tailed dataset comprising K categories, where the number of samples for each category k is denoted as n_k, this "difficulty coefficient" can be defined as

$$D = D_L + \sum_{k=1}^{K} d_k \tag{1}$$

where D_L represents the harmful extent of the long-tailed distribution inflicting upon the model generalization capability

$$D_L = W \cdot D_F \tag{2}$$

$$W = \sum_{k=1}^{K} W_k \tag{3}$$

$$W_K = \frac{1}{n_k}. \tag{4}$$

Intuitively, D_L is positively correlated with the long-tailed distribution degree of the training data. D_F denotes the impact of the framework on the difficulty of classifying long-tail data.

For classification tasks, building a suitable feature mapping space is an essential step to achieve better classification performance, since it is critical for capturing the interclass differences among samples. As shown in (5), a distance coefficient d_k is defined to abstract the inherent classification difficulty between mapping features of class k and other categories.

d_k represents the inherent classification difficulty of the kth category, which can be measured through interclass dissimilarity

$$d_k = \frac{\sum_{i=1}^{n_k} \sum_{j=1, j\neq k}^{K} s(E(x_{ki}), E(x_j))}{n_k \cdot (K-1)} \tag{5}$$

where x_{ki} represents the ith sample feature of the kth category; E denotes the feature extractor; $s(E(x_{ki}), E(x_j))$ calculates the similarity level between the features of sample x_{ki} and all samples in category j. A smaller $s(E(x_{ki}), E(x_j))$ indicates a greater distinction between different categories. Limited samples in certain tail categories prevent the model from fully harnessing the information within these categories. Consequently, there is a reduction in information saturation within the features generated by the extractor, impacting the model's performance and increasing the level of classification difficulty. In summary, the difficulty coefficient D serves as an intuitive abstraction of the challenges in long-tailed classification. A large D implies a difficult classification task, while a small D indicates a simpler task. The algorithm's objective is to optimize the framework F and feature extractor E to minimize the difficulty coefficient D. Consequently, existing RSIC methods tend to perform poorly when dealing with long-tailed data.

To solve the above issues, in recent years, researchers have carried out a series of studies on long-tailed RSIC. Extensive studies can be divided into three strategies: class rebalancing, framework refinement, and representation improvement. The class rebalancing of long-tailed data typically involves oversampling the tail classes and undersampling the head classes. Oversampling [10], [11], [12], [13] increases the amount of data in the tail classes, while undersampling [14] reduces the amount of data in the head classes, thereby rebalancing the data distribution between the head and tail classes. In addition to oversampling and under-sampling, Feng et al. [15] proposed monitoring the model training process across different classes using the mean classification prediction score and utilized this metric to guide the sampling rates for diverse classes. Kang et al. [16] proposed four different sampling methods including instance-balanced sampling, class-balanced sampling, square root sampling, and progressively-balanced sampling. Class rebalancing is aimed at reducing the impact of n_k on D_L.

The framework refinement primarily refers to decoupled learning, separating the learning process into representation learning and classifier training to mitigate the damage caused by a class imbalance in the model. Kang et al. [16] used a straightforward approach that decouples representation and classification and found that it is possible to outperform carefully designed losses, sampling strategies, and even complex modules with memory. Li et al. [17] proposed a novel self-distillation scheme to improve decoupled training, generating robust soft labels from both data and label domains, effectively transferring knowledge from the head class to the tail class. Zhang and Xiang [18] proposed a new method, gradual balanced loss and adaptive feature generator (GLAG), which can combine with different decoupled training methods to improve the original performance. The purpose of framework refinement is to decrease D_L through F. The limitation of class rebalancing is that it fundamentally does not improve the sufficient information for tail class data, and decoupled learning can also result in incompatibility between feature extractors and classifiers.

Representation improvement methods are designed to enhance the ability of feature extractors, which are essential approaches to solve the long-tailed issue. Cost-sensitive learning improves the representational capability of the feature extractor by focusing on the loss function. Cui et al. [19]

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designed a reweighting scheme that uses the effective number of samples for each class to rebalance the loss, thereby yielding a class-balanced loss. Lin et al. [20] proposed focal loss to inversely reweight classes using prediction probabilities, which assign higher weights to the harder tail classes but lower weights to the easier head classes. Zang et al. [21] explored the model classification loss on a balanced meta-validation set as a score, adapting the sampling rate for different classes to prioritize increased sampling for under-represented tail classes. Wang et al. [22] employed output confidence to determine whether to suppress the gradient for a negative label, addressing the issue that simply down-weighting negative gradients may harm the discriminative abilities of deep models. Wang et al. [23] proposed Seesaw loss to dynamically rebalance gradients of positive and negative samples for each category, with two complementary factors, i.e., mitigation factor and compensation factor. Zhang et al. [24] utilized statistics from the entire batch rather than the instance level, thereby reducing the bias caused by imbalances in data quantity across different classes. Liu et al. [25] developed an integrated OLTR representation learning algorithm including dynamic meta-embedding and modulated attention module, which effectively transfer knowledge from head class to tail classes and discriminate between them. In summary, existing representation improvement methods aim to optimize the feature extractor E to reduce inherent classification difficulty d_k, and the effectiveness still has significant room for improvement.

Addressing the issues with existing methods, this article introduces a dynamic decoupling (DD) and multiobjective optimization for long-tailed RSIC, namely, DECOR. To mitigate the mutual influence between the representation and the classification stage, the training phase of the feature extractor and classifier is executed as a dynamic decoupled approach to improve feature learning and classifier training. Due to the incompatibility that arises from complete decoupling between the feature extractor and the classifier, the performance of both components is ensured through the proposed dynamic decoupled training scheme, thereby enhancing the D_F of the model framework. To optimize the feature extractor E, this article adopts a multiobjective optimization framework (MOOF), synchronously conducting joint training for supervised contrastive learning with learnable feature centroid (LFC) and masked world knowledge learning (WKL), which enables the feature extractor to acquire better image representations. In supervised contrastive learning, an LFC is trained for each category to increase interclass dissimilarity and intraclass similarity. This helps establish an explicit connection between the feature extractor and the classifier while simultaneously reducing sample overlap among head, middle, and tail classes. When humans acquire background knowledge about how the world operates, it is done through observation and in a task-independent, unsupervised manner, and this accumulated knowledge may constitute the foundation of what is commonly referred to as common sense. Self-supervised learning (SSL) allows us to train a model to learn better representations of various inputs in a task-independent manner. Since SSL utilizes only unlabeled data, any training dataset can be leveraged to enable the model to learn more robust and comprehensive input representations, thereby introducing common-sense world knowledge into the model. In order to effectively acquire world knowledge, the features extracted by the feature extractor undergo dual random masked enhance (DRME) projection. Furthermore, a lightweight optimization fine-tuning (LOFT) technique is employed with the aim of parameter fine-tuning. The algorithm proposed in this article achieved an average improvement of 6.61% on the artificially generated NWPU-LT remote sensing dataset, 17.36% on the artificially generated AID-LT remote sensing dataset, and 13.62% improvement on our self-constructed long-tailed dataset BIT-AFGR50-LT.

The main contributions of this article are summarized as follows.

1) A DD framework is proposed that allows the model to make a better representation of learning and classifier learning, and also ensure the compatibility of the feature extractor with the classifier.
2) MOOF is proposed to make a better representation of learning including supervised contrastive learning with LFC and self-supervised contrastive learning. LFCs enable a more explicit connection between the feature extractor and the classifier. Self-supervised contrastive learning provides the model with contextual knowledge about the world.
3) A LOFT is employed for the goal of maximum performance with minimal intervention.
4) A high-spatial-resolution remote sensing image long-tailed dataset containing 50 classes of objects has been constructed by ourselves and will be made publicly available to other researchers. The self-built BIT-AFGR50 is available at https://github.com/wgqqgw/BIT-KTYG-AFGR. We validated the effectiveness of our algorithm on three datasets, including the artificially induced NWPU-LT dataset, the artificially induced AID-LT dataset, and our self-constructed BIT-AFGR50-LT dataset.

II. Related Work

In this section, some works related to long-tailed classification contrastive learning and decoupling learning will be reviewed.

A. Long-Tailed Classification

Long-tailed distribution is a challenge for classification tasks, where a small portion of classes have massive samples but the others have a few samples.

Most algorithms are proposed to solve the issue. Many algorithms are proposed to solve the issue. For example, Wang et al. [26] proposed a new long-tailed classifier called routing diverse experts (RIDEs), which reduces the model variance with multiple experts, reduces the model bias with a distribution-aware diversity loss, reduces the computational cost with a dynamic expert routing module. Wang et al. [27] proposed a novel hybrid network structure composed of a supervised contrastive loss to learn image representations and

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a cross-entropy loss to learn classifiers, where the learning is progressively transited from feature learning to classifier learning. Cui et al. [28] proposed parametric contrastive learning (PaCo) to tackle long-tailed recognition, which introduces a set of parametric class-wise learnable centers to rebalance from an optimization perspective. Zhao et al. [29] first designed a novel adaptive logit adjustment (ALA) loss, which contains an instance-specific adjusting term that is adaptive to the logit of each sample and can make the model pay more attention to hard samples and then propose an end-to-end learnable mixture-of-experts (MoEs) network, with a multiexpert module and an instance-aware routing module, which can dynamically integrate the results of multiple experts and boost the performance. Cui et al. [19] designed a class-balanced loss that uses the effective number of samples for each class to rebalance the loss.

B. Contrastive Learning

Contrastive learning has recently become an important component in SSL methods for computer vision and other domains, which aims at grouping similar samples closer and diverse samples far from each other.

He et al. [30] presented momentum contrast (MoCo) for unsupervised visual representation learning, which builds a dynamic dictionary with a queue and a moving-averaged encoder so that a large and consistent dictionary can be built on the fly to facilitate contrastive unsupervised learning. Chen et al. [31] proposed a simple framework for contrastive learning of visual representations (SimCLRs) without requiring specialized architectures or a memory bank. Caron et al. [32] proposed an online algorithm, SwAV, that simultaneously clusters the data while enforcing consistency between cluster assignments produced for different augmentations (or "views") of the same image and then proposed a new data augmentation strategy, multicrop, that uses a mix of views with different resolutions in place of two full-resolution views, without increasing the memory or computing requirements. Grill et al. [33] proposed bootstrap your own latent (BYOL), a new approach to self-supervised image representation learning, which relies on two neural networks, referred to as online and target networks, that interact and learn from each other. Chen and He [34] proposed a simple Siamese network for self-supervised representation learning, even not using negative sample pairs, large batches, and momentum encoders, and show that a stop-gradient operation plays an essential role in preventing collapse. Unlike self-supervised contrastive learning, supervised contrastive learning leverages supervised information from labels. Khosla et al. [35] extended the self-supervised batch contrastive approach to the fully-supervised setting to effectively use label information. Clusters of points belonging to the same class are pulled together in embedding space by supervised contrastive learning, while simultaneously pushing apart clusters of samples from different classes.

C. Decouple Learning

Coupling refers to the relationship of two components that can collaborate. If A and B can collaborate, then the relationship between A and B is coupling. Decoupling, on the other hand, is the dissolution of this collaborative relationship so that they become two unrelated parts.

Decoupling [36], [37] learning decouples the model into representation learning and classifier training. Kang et al. [16] proposed the pioneering work to introduce the two-stage training scheme, which adopted different sampling strategies in representation learning and then fixed the feature extractor to check various classifier strategies in the second stage. Desai et al. [38] proposed decoupled training for devil in the tails (DT2) based on an iterative decoupled training scheme, which employs a novel sampling approach, alternating class-balanced sampling (ACBS), to capture the interplay between the long-tailed entity and predicate distributions of visual relations. Zhang et al. [39] developed an adaptive calibration function that enables us to adjust the classification scores for each data point and then introduce a generalized reweight method in the two-stage learning to balance the class prior, which provides a flexible and unified solution to diverse scenarios in visual recognition tasks. Zhong et al. [40] proposed a mixup shifted label-aware smoothing model (MiSLAS), which handled different degrees of over confidence for classes by label-aware smoothing and dealt with the dataset between two stages of bias due to different samplers by shifted batch normalization in decoupling framework. Zhang and Xiang [18] GLAG, which can combine with different decoupled training methods to improve the original performance.

III. Method

A. Preliminaries

1) Self-Supervised Contrastive Learning: Self-supervised contrastive learning holds a critically significant role in the realm of computer vision, offering a solution to the challenge of acquiring meaningful visual representations without the need for supervision, such as labels. With the development of contrastive learning, discriminative methods based on contrastive learning in the latent space have recently shown great potential for advanced results. SimCLR is one of the important models of SSL, and the architecture of which can be simply divided into three parts: data augmentation $\mathrm{aug}(\cdot)$, encoder $f(\cdot)$, and projector $g(\cdot)$. The role of the encoder is to extract the feature from the augmented image. The role of the projector is to map features to the space where contrastive loss is applied. The process of SimCLR can be described as follows: first, the dataset is randomly divided into several minibatches, and the number of samples in each small batch is N. For a set of N randomly sampled sample/label pairs $\{x_k, y_k\}_{k=1,2,\ldots,N}$, two different data augmentations are done on each image to get a set of $2N$ sample/label pairs $\{\hat{x}_k, \hat{y}_k\}_{k=1,2,\ldots,2N}$. Two different transformations of the same sample are defined as positive pairs, and the transformed samples between different samples are defined as negative pairs. Thus given a positive pair, the other $2(N-1)$ augmented examples within a minibatch are treated as negative examples. The transformed images are passed through two encoders $f(\cdot)$ with shared weights, which are structured as convolutional

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neural networks, such as the ResNet [41] series, or other convolutional neural network models such as Visual Geometry Group (VGG) [42] and GoogleNet [43]. The feature obtained by encoding is passed through a projector $g(\cdot)$ named multilayer perceptron (MLP) with a single hidden layer to project the image features into a D_s-dimensional projection space. The obtained projection vectors are normalized and put into a self-supervised contrastive loss

$$L^{\text{self}} = \sum_{i \in I} L_i^{\text{self}} = -\sum_{i \in I} \log \frac{\exp(z_i \cdot z_j/\tau)}{\sum_{l \in G} \exp(z_i \cdot z_l/\tau)} \tag{6}$$

where $z_i = \text{norm}(g(f(\text{aug}(x_i)))) \in \mathcal{R}^D$, $\text{norm}(\cdot)$ is the ℓ_2 normalization, $I = \{1, 2, 3, \ldots, 2N\}$, and j is the index of the other augmented sample originating from the same source sample. $G = \{1, 2, 3, \ldots, i-1, i+1, \ldots, 2N\}$, τ is a temperature parameter.

2) Supervised Contrastive Learning: Supervised contrastive learning has the same architecture as self-supervised contrastive learning and consists of three parts: data augmentation $\text{aug}(\cdot)$, encoder $f(\cdot)$, and projector $g(\cdot)$. The difference between them lies in the division of positive and negative pairs. In supervised contrastive learning, samples belonging to different transformations of the same category are considered positive pairs, and samples of different categories are considered negative pairs. The process of supervised contrastive learning is also consistent with self-supervised contrastive learning. The samples with different transformations are first passed through the encoder to obtain the feature, and then the projector maps the feature into the space where contrastive loss is applied. The projection vector is normalized and put into supervised contrastive learning loss

$$\begin{aligned} L^{\text{SupCon}} &= \sum_{i \in I} L_i^{\text{SupCon}} \\ &= \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p/\tau)}{\sum_{n \in G} \exp(z_i \cdot z_n/\tau)} \end{aligned} \tag{7}$$

where $z_i = \text{norm}(g(f(\text{aug}(x_i)))) \in \mathcal{R}^D$, $\text{norm}(\cdot)$ is the ℓ_2 normalization, $I = \{1, 2, 3, \ldots, 2N\}$, $G = \{1, 2, 3, \ldots, i-1, i+1, \ldots, 2N\}$, $P(i) = \{p \in G | y_p = y_i\}$, p is the index of a positive sample of x_i different from i, $|P(i)|$ is the number of positive samples of class y_i in a minibanch, and τ is a temperature parameter.

B. Method Overview

In the training stage, the proposed DECOR is a DD framework with dynamic fine-tuning, which consists of representation learning and classifier learning. The parameters of the feature extractor are optimized through dynamic fine-tuning. Specifically, as the training iteration increases, the learning focus of the model shifts from representation learning to classifier learning, which can achieve optimal compatibility for both feature extractor and classifier. In addition, in order to introduce unlabeled world knowledge and fine-tune the classifier with labeled task-specific data, a linear adapter is applied to the classifier module as shown in Fig. 2. Next, the training details of the proposed model will be delineated.

In this article, the proposed DD dynamic fine-tuning framework prioritizes representation learning in the initial stages of training, focusing on enhancing the representation capability of the feature extractor. In this stage, a MOOF is introduced to elevate representation learning, in which supervised contrastive learning with feature centroids and WKL are concurrently integrated into the training process, co-optimizing the feature extractor. It is noteworthy that we do not adopt a serialized training strategy, wherein supervised contrastive learning with feature centroids precedes WKL. Concretely, the learned feature centroids from the centroid branch and the sample features from the feature branch are incorporated into the supervised contrastive loss with LFC, denoted as $L^{\text{centroidSC}}$, to bolster the representational prowess of the feature extractor. Simultaneously, unlabeled images are used to incorporate a wealth of world knowledge, addressing the insufficient representation of tail classes by self-supervised contrastive learning. Through the approach of joint collaborative training, our model is capable of globally enhancing representation capabilities and significantly acquiring world-centric common knowledge, effectively addressing the challenge of few-shot learning on tail classes. The advantage of this cooperative interaction lies in the model's simultaneous benefit from both task-specific data and world knowledge, leading to more pronounced performance improvements when handling multiobjective tasks.

In the second half of the DD training, i.e., the classifier learning stage, class-balanced resampling is initially adopted to obtain balanced data from the long-tailed dataset. The classifier will remodel a balanced decision boundary under balanced data conditions, which can take advantage of the intraclass compactness and interclass separability of the feature extractor obtained by the MOOF. In addition, a LOFT is proposed with the aim of parameter fine-tuning.

In the method, the remote sensing image dataset is divided into three subsets: $\mathcal{L}$ and $\mathcal{U}$ training sets for training purposes, and the $\mathcal{T}$ test set for testing purposes. During the representation learning stage, $\mathcal{L}$, which is a long-tail dataset, is used for supervised contrastive learning with LFC, while $\mathcal{U}$, an unlabeled dataset, is employed for SSL. The balanced data obtained through class-balanced sampling from dataset $\mathcal{L}$ is used for the classifier learning stage. The test set $\mathcal{T}$ is a balanced dataset used to evaluate the trained model.

C. Multiobjective Optimization Framework

A MOOF can help models achieve better representation capability through supervised contrastive learning with LFC and WKL.

In supervised contrastive learning, the concept of LFC is introduced, which serves to pull samples from the same class closer together while pushing samples from different classes apart. Each sample only needs to be contrasted with LFCs from all categories, reducing sample overlap among head, middle, and tail classes. In the MOOF, cosine similarity is used to measure the similarity between two projected feature vectors. Assuming v_1 and v_2 are unit vectors, i.e., $\|v_1\| = \|v_2\| = 1$, then the cosine similarity is defined as

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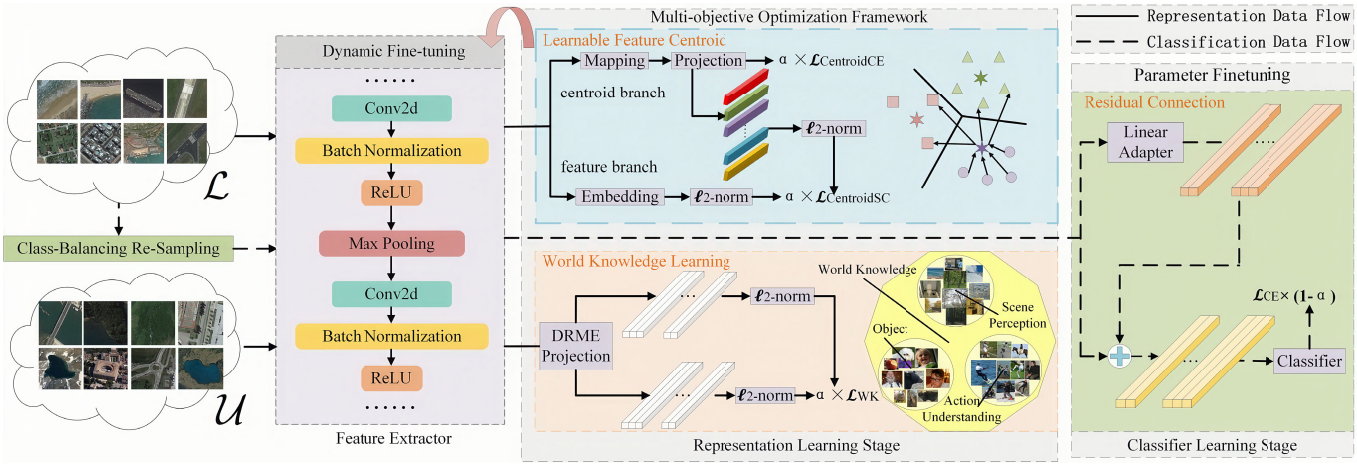


Fig. 2. Overall architecture of DECOR. By employing the DD framework, our model training pivots from the seamless progression of representation learning to classifier learning. The parameters of the feature extractor undergo dynamic fine-tuning through a multiobjective optimization strategy that synchronously combines supervised contrastive learning with LFC and WKL. Furthermore, to accommodate the world knowledge embedded in the multiobjective optimization strategy, we execute parameter fine-tuning on the classifier module.

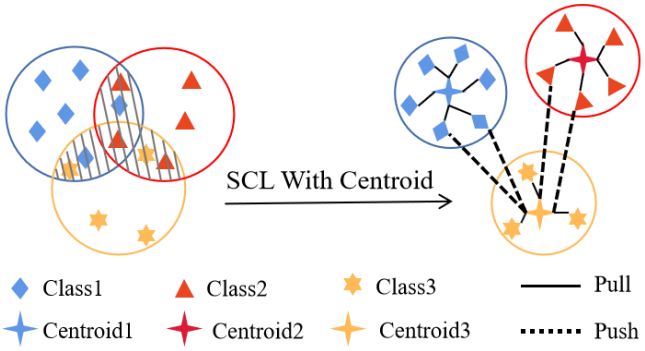


Fig. 3. Supervised contrastive learning with learnable centroid.

$\cos\theta = v_1{}^T v_2$. The larger the value of $\cos\theta$, the higher the similarity between the two vectors. In the centroid branch, the probability p_i is calculated using the softmax function with the formula $p_i = \text{softmax}(W^T r_i)$, where elements of p_i signify the likelihood of sample x_i belonging to respective classes. Here, $r_i \in \mathcal{R}^{D_f}$ denotes the projection vector of a sample x_i after feature extraction and mapping, W is the $(D_f, D_{\text{classes}})$ dimensional weight matrix of projection in the centroid branch. This allows us to comprehend the operation of projection as measuring the cosine similarity between the r_i and the columns of the weight matrix of the projection. Therefore, each column of the weight matrix W of projection serves as the learnable feature centriod for a class. By examining the indices of the probability vector p_i, the first and last columns of W are identified as the feature centroids for the initial and final classes, respectively. In addition, the LFC is fine-tuned following the equation, where p_i is the above-mentioned dynamic classification likelihood of projected sample x_i

$$L^{\text{CentroidCE}} = \sum_{i\in I} L_i^{\text{CentroidCE}} = \sum_{i\in I} -p_i \log p_i. \tag{8}$$

The parameters were updated via backpropagation, resulting in the adaptable feature centroids. The supervised contrastive learning with LFC is shown in Fig. 3. This allows the feature centroid to pull samples of the same class together and push away samples from different categories, thus obtaining a feature extractor with excellent performance.

As shown in Fig. 2, the supervised contrastive learning with LFC consists of two branches, which are the centroid branch and the feature branch. The centroid branch is used to obtain the LFC, and the feature branch is a process in which the samples in the minibatch are passed through the feature extraction and the embedding in turn to obtain the representation vector z_i. The pairs of the sample representation vector and the centroid of this sample category are considered positive pairs, and the pairs of this sample representation vector and the feature centroid of other categories are considered negative pairs. After normalizing the representation vector and the centroid, they are put into the LFC-supervised contrastive loss

$$\begin{aligned} L^{\text{CentroidSC}} &= \sum_{i\in I} L_i^{\text{CentroidSC}} \\ &= \sum_{i\in I} -\log \frac{\exp(z_i \cdot P_{y_i}/\tau)}{\sum_{j=1, j\neq y_i}^{C} \exp(z_i \cdot P_j/\tau)} \end{aligned} \tag{9}$$

where z_i is the normalized representation vector of x_i, P_{y_i} is the feature centroid of y_i class, $I = \{1, 2, 3, \ldots, 2N\}$, N is the batch size of the minibatch, C is the number of categories in the dataset, and τ is a temperature parameter.

WKL is another crucial component of the MOOF. In this segment, the model incorporates a vast array of unlabeled images from the natural world as world knowledge through self-supervised contrastive learning. This integration enhances the model's common sense representation capability significantly. Self-supervised contrastive learning provides models with common-sense world knowledge in areas such as object recognition, scene understanding, and action comprehension, leveraging background knowledge of the world to address the limitations of insufficient data. Specifically, feature vectors generated from the same sample through a dual random masked projection are labeled as positive pairs, while pairs of samples generated from different samples by the random masked projection are considered negative pairs. As shown in Fig. 2, the dual random masked enhanced projection serves

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as a key component of WKL. This technique operates on feature maps, generating a masked enhanced feature (mef) vector by randomly masking feature points on the feature map (fm) at various proportions, and then followed by enhanced projection. Unlike common pixel-based masking methods, where the model attempts to fill in missing information at each pixel, overly focusing on irrelevant details rather than capturing high-level predictable concepts, masking features make it easier for the model to grasp abstract representations of advanced information in the masked portion. The formula for this process can be expressed as follows:

$$\begin{aligned} \text{mef} &= \text{DRMEP}(fm, \text{rate}) \\ &= \text{EP}(\text{RM}(fm, \text{rate})) \end{aligned} \tag{10}$$

where RM stands for random mask, and EP stands for enhanced projection. WKL endeavors to learn and comprehend obscured feature-level information and high-level abstractions within images. This imparts the model with common-sense world knowledge, significantly enhancing the model's generalization capabilities. So the world knowledge loss function is

$$\begin{aligned} L^{\text{WK}} &= \sum_{i \in I} L_i^{\text{WK}} \\ &= -\sum_{i \in I} \log \frac{\exp(\text{mef}_i \cdot \text{mef}_j/\tau)}{\sum_{l \in I, l \neq i} \exp(\text{mef}_i \cdot \text{mef}_l/\tau)} \end{aligned} \tag{11}$$

where mef_i represents the projected feature vector obtained from sample x_i after passing through the feature extractor and dual random masked enhanced projection, j is the index of another augmented sample of x_i. $I = \{1, 2, 3, \ldots, 2N\}$, j is the index of the other augmented sample originating from the same source sample. τ is a temperature parameter. So the loss function for feature extraction is

$$L^{\text{Extractor}} = a \cdot L^{\text{WK}} + b \cdot L^{\text{CentroidSC}} + c \cdot L^{\text{CentroidCE}}. \tag{12}$$

D. Class Balancing Resampling and LOFT

To prevent extremely unbalanced data from affecting better classifier construction, the class balancing resampling is proposed to get a balanced classification decision boundary. Class balancing resampling first samples the categories from the original long-tailed dataset in equal probability and then randomly samples a data point from the selected class. Due to the limited number of samples in tail categories, we intend to utilize lightweight fine-tuning techniques to enhance the model's performance on tail classes, aiming to achieve the goal of minimal intervention for maximum performance. The samples are extracted by the backbone's features to get feature f. Assuming the weight matrix and bias of LOFT is $W \in \mathcal{R}^{D_c \times D_c}$ and $b \in \mathcal{R}^{D_c}$, so the image feature can be represented as

$$f' = \lambda \cdot (W^T f + b) + (1 - \lambda) \cdot f \tag{13}$$

where λ indicates the residual factor to dynamically combine fine-tuned image features of the classifier learning phase with the original image features of the representation learning phase. The LOFT serves as a bridge between world knowledge and task-specific data. Through fine-tuning, it adapts common sense features to the downstream tasks, enhancing the classifier's performance. The loss function of classifier is $L^{\text{classifier}}$

$$L^{\text{Clasiifier}} = \sum_{i \in I} L_i^{\text{classifier}} = \sum_{i \in I} -p_i' \log p_i' \tag{14}$$

where $p_i' = \text{softmax}(W_c^T f_i')$ and W_c is the weight of classifier.

Algorithm 1 Pseudocode of DECOR

Input: Training epoch T, Batchsize $\{N_l, N_w, N_c\}$, Labeled data $\mathcal{L}$, Unlabeled data $\mathcal{U}$, Encoder e, Embedding g_{MLP}, Mapping g_c, projection g_p, Dual Random Masked Enhanced Projection $DRMEP$, Random Mask Rate $\{rate_1, rate2\}$ Augmentation $\{aug_1, aug_2\}$, Constant $\{a, b, c\}$

for *t=1,* $2, 3, \ldots, T$ **do**
  for *randomly sampled minibatches* $\{x_{li}\}_{i=1}^{N_l} \in \mathcal{L}$, $\{x_{ui} \in\}_{i=1}^{N_w} \in \mathcal{U}$ *and* $\{x_{cbsi}\}_{i=1}^{N_c} \in \mathcal{L}$ **do**
    for *ALL* $i \in \{1,2,\ldots, N_l\}$ **do**
      $z_{li1} = g_{MLP}(f(aug_1(x_{li})))$
      $z_{li2} = g_p(g_c(f(aug_2(x_{li}))))$
    end
    Get the feature centoid from the projection
    Compute loss: $L_{CentroidSC}$
    Compute loss: $L_{CentroidCE}$
    for *ALL* $i \in \{1, 2, \ldots, N_w\}$ **do**
      $mef_{ui1} = DRMEP(f(aug_1(x_{ui})), rate_1)$
      $mef_{ui2} = DRMEP(f(aug_2(x_{ui})), rate_2)$
    end
    Compute loss: L_{WK}
    # Dynamic finetuning for representation
    for *ALL* $i \in \{1, 2, \ldots, N_c\}$ **do**
      $z_i = f(x_{cbsi})$
      $z_i' = z_i + Adapter(z_i)$
      $logits_i = classifier(z_i')$
    end
    Compute loss: $L_{Classifier}$
    # Parameter finetuning for classification
    α = t/T
    Compute the final loss:
    $Loss = (1 - \alpha) \cdot (a \cdot L_{WK} + b \cdot L_{CentroidSC} + c \cdot L_{CentroidCE}) + \alpha \cdot (L_{Classifier})$
    Update the network;
  end
end

E. DD Learning

In order to obtain optimal network parameters, a DD learning strategy with dynamic fine-tuning is proposed in this article. Specifically, the feature extractor and classifier are trained separately to prevent the influence of the classifier on the feature extractor during backpropagation since the backpropagation process is a chained process, and also to ensure that good features promote a good classifier at the same

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time. However, completely decoupling the feature extractor and classifier affects the compatibility of the feature extractor and classifier and affects the classification performance, so a DD approach is proposed to make the training process gradually transition from the feature extractor to the classifier. Utilizing the hyperparameter α, the ultimate loss function is derived

$$\text{Loss} = (1-\alpha) \times L^{\text{Extractor}} + \alpha \times L^{\text{Classifier}} \tag{15}$$

$$\alpha = (\text{ep}/\text{eps})^{\beta} \tag{16}$$

where $L^{\text{Extractor}}$ is the loss function of the feature extractor, $L^{\text{Classifier}}$ is the loss function of the classifier, ep is the current training epoch, and eps is the total number of training epochs.

IV. Experiments

A. Datasets

To validate the effectiveness of the proposed DECOR, we conducted experiments on several remote RSIC datasets with long-tailed distribution. First, to verify the availability of the proposed DECOR in dealing with long-tail distributed remote sensing data, we sample each dataset in an unbalanced way so that it conforms to the long-tailed distribution. As shown in Tables I and II, the datasets are manually divided into many shots, medium shots, and few shots

$$N_i = N_{\max} \times \text{imb}^{\left(\frac{i}{C-1}\right)}. \tag{17}$$

Consequently, the long-tailed datasets can be constructed by sampling each dataset with three different imbalance factors according to (17), where N_i is the number of samples in category i, $N_{\max}$ is the number of samples in the category containing the most samples, and imb is the imbalance factor which is used to indicate the ratio of sample volumes in the head class to the tail class after sampling. Specifically, the imbalance factors are set to 0.01, 0.02, and 0.1. C is the total number of categories, and $i \in \{1, 2, 3, \ldots, C\}$. Details of the constructed long-tailed datasets are elaborated below.

1) NWPU-LT: The NWPU-RESISC45 [44] is a typical publicly available dataset for remote sensing scene classification which is proposed by the researchers of Northwestern Polytechnical University (NWPU). It contains 45 scene classes: airplane, airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. Each category of the dataset has 700 images with 256 × 256 pixels, for a total of 31 500; 100 images per category are randomly selected as the test set. The remaining 600 images from each category are applied to generate the NWPU-LT according to (17), where $C = 45$, $N_{\max} = 600$, and imb are set to 0.01, 0.02, and 0.1, respectively. Detailed information about different classes is provided in Table I.

TABLE I
Detailed Information About NWPU-LT

Classname	Training Samples			Classname	Training Samples		
	imb=0.01	*imb*=0.02	*imb*=0.1		*imb*=0.01	*imb*=0.02	*imb*=0.1
Overpass	600	600	600	Intersection	540	548	569
Rectangular Farmland	486	502	540	Desert	438	459	512
Wetland	394	420	486	Industrial Area	355	384	461
Circular Farmland	320	351	438	Terrace	288	322	415
Harbor	259	294	394	Medium Residential	233	269	374
Railway	210	246	355	Church	189	225	337
Palace	170	206	320	Roundabout	153	188	303
Baseball diamond	138	172	288	Cloud	124	158	273
Tennis Court	112	144	259	Chaparral	101	132	246
Parking Lot	91	121	233	Island	82	110	221
Runway	73	101	210	Airport	66	92	199
Stadium	60	84	189	Freeway	54	77	180
Forest	48	71	170	Golf Course	43	64	162
Meadow	39	59	153	Basketball Court	35	54	146
Beach	32	49	138	Ground Track Field	28	45	131
Lake	25	41	124	Commercial Area	23	38	118
Mobile Home Park	21	34	112	Ship	18	31	106
Railway Station	17	29	101	Snowberg	15	26	96
Dense Residential	13	24	91	Sea Ice	12	22	86
Airplane	11	20	82	Bridge	10	18	77
Storage Tank	9	17	73	Mountain	8	15	70
Sparse Residential	7	14	66	Thermal Power Station	6	13	63
River	6	12	60	Total	5962	6901	10627

TABLE II
Detailed Information About AID-LT

Classname	Training Samples			Classname	Training Samples		
	imb=0.01	*imb*=0.02	*imb*=0.1		*imb*=0.01	*imb*=0.02	*imb*=0.1
Airport	100	100	100	Bareland	85	87	92
Baseball Field	72	76	85	Beach	62	66	78
Bridge	52	58	72	Center Area	45	50	67
Church	38	44	62	Commercial	32	38	57
Dense Residential	28	33	52	Desert	23	29	48
Farmland	20	25	45	Forest	17	22	41
Industrial	14	19	38	Meadow	12	17	35
Medium Residential	10	15	32	Mountain	9	13	30
Park	7	11	28	Parking Lot	6	10	25
Playground	5	8	23	Pound	4	7	22
Port	3	6	20	Railway Station	3	5	18
Resort	2	5	17	River	2	4	16
School	1	3	14	Sparse Residential	1	3	13
Square	1	2	12	Stadium	1	2	11
Storage Tanks	1	2	10	Viaduct	1	2	10

2) AID-LT: The AID [45] is another benchmark dataset for optical remote sensing scene classification jointly proposed by the Huazhong University of Science and Technology and Wuhan University in 2017, which consists of 30 scene classes: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. Each category contains around 220–420 images with 600 × 600 pixels. Fifty images in each category are selected as the test set and the remaining images of the AID dataset are generated as the AID-LT according to (17), where $C = 30$, $N_{\max} = 100$, and imb are set to 0.01, 0.02, and 0.1, respectively. Detailed information about AID-LT is listed in Table II.

3) BIT-AFGR50-LT: The BIT-AFGR50-LT is our self-built long-tailed fine-grained RSIC dataset, which is constructed based on our previous work [46] and released at https://github.com/wgqqgw/BIT-KTYG-AFGR. It contains 36 278 image instances with 50 categories. Similar to the number of aircraft targets in each category that exists in reality, the sample volume of each category in BIT-AFGR50-LT is not balanced and has a long-tailed distribution, as shown in Fig. 4. BIT-AFGR50-LT simulates the data distributions in real environments, which facilitates researchers

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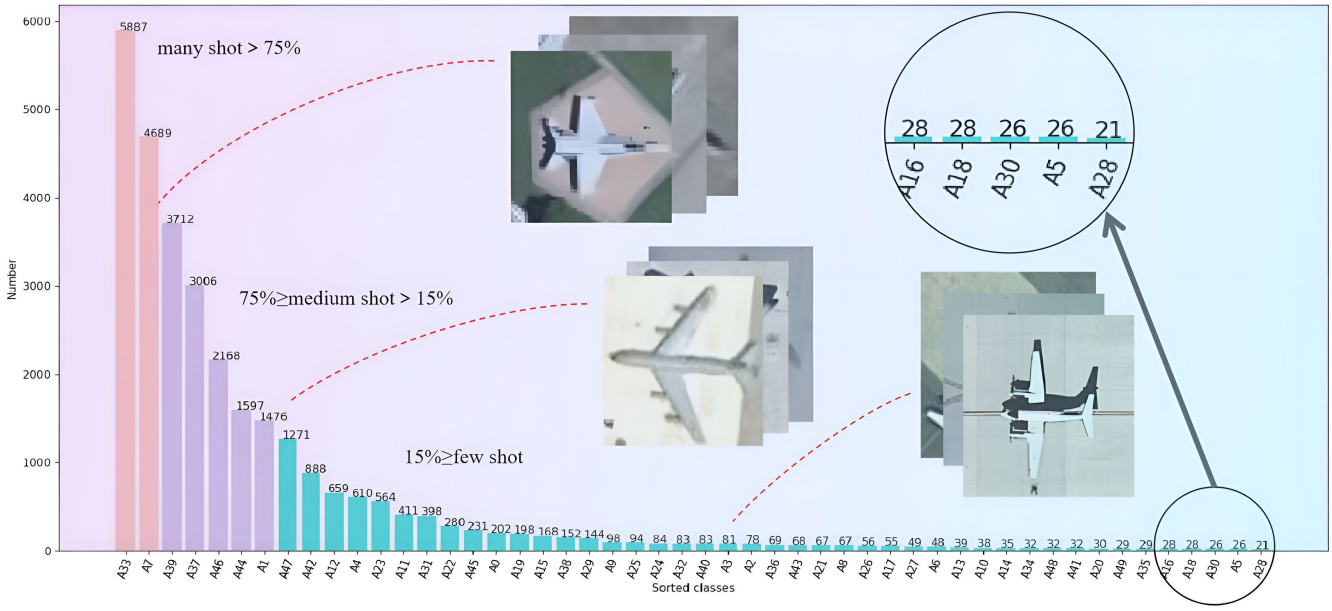


Fig. 4. Distribution of sample quantities among different categories in the BIT-AFGR50-LT dataset follows a long-tail distribution.

to carry out algorithmic studies oriented to long-tailed data distributions. Details are shown in Fig. 4. In contrast to NWPU-LT and AID-LT, which primarily emphasize scene classification, BIT-AFGR50-LT is a long-tail dataset that pertains to fine-grained categories of aircraft. This aspect aids in showcasing the formidable generalization ability of our algorithm. Consequently, we have opted for BIT-AFGR50-LT as the benchmark for testing our algorithm on a long-tail dataset.

B. Implementation Details and Evaluation Metrics

1) Implementation Details for NWPU-LT: We implement our DECOR with Pytorch on 2 Tesla V100 GPUs for 100 epochs. The input images are transformed to 224 × 224. Stochastic gradient descent (SGD) [54] with momentum 0.9 and weight decay 1×10^{-4} is selected as the optimizer. In addition, Resnet-50 [41] is adopted as the backbone, and the batch size is set to 64 with an initial learning rate of 0.005. The learning rate is decayed by a factor of 10 in the 50th epoch and 80th epoch, respectively. The temperature of supervised contrastive learning with LFC is fixed at 0.6, and the temperature of WKL is fixed at 0.5. Hyperparameters {β, a, b, c} in (15) and (12) are set to {1, 0.3, 0.5, 0.5}, respectively.

2) Implementation Details for AID-LT: We use SGD optimizer with momentum 0.6 and weight decay 1×10^{-4}. Resnet-50 [41] is adopted as the backbone and the batch size is set to 64 with the initial learning rate of 0.01. The learning rate is decayed by a factor of 10 in the 45th epoch and 75th epoch, respectively. The temperature of supervised contrastive learning with LFC is fixed to be 0.6, and the temperature of WKL is fixed to be 0.5. Hyperparameters {β, a, b, c} in (15) and (12) are set to {2, 0.8, 0.5, 0.5}, respectively.

3) Implementation Details for BIT-AFGR50-LT: We use SGD optimizer with momentum 0.9 and weight decay 1×10^{-4}. The core architecture is ResNet-50 [41], serving as the backbone for our model. The batch size is 64, with an initial learning rate set to 0.05. The learning rate is decayed by a factor of 10 in the 45th epoch and 80th epoch, respectively. The temperature of supervised contrastive learning with LFC is fixed to be 0.6, and the temperature of WKL is fixed to be 0.5. Hyperparameters {β, a, b, c} in (15) and (12) are set to {1, 0.8, 0.5, 0.5}, respectively.

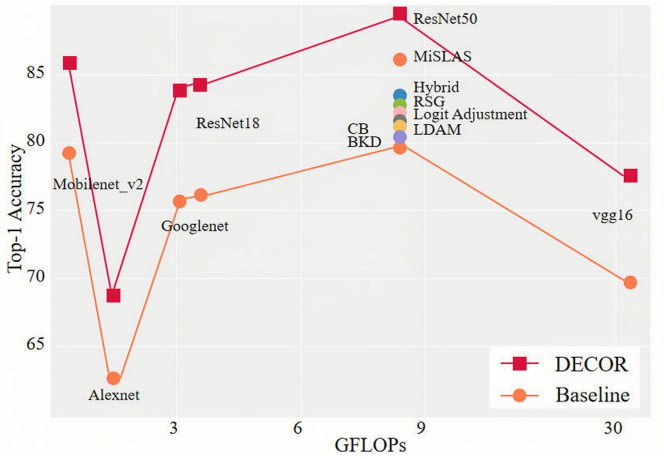


Fig. 5. Complexity/accuracy tradeoffs. We illustrate the tradeoffs between model complexity and top-1 accuracy (%) for both the baseline and our model on the NWPU-LT dataset with an imbalance rate of 0.01.

4) Evaluation Metrics:

a) Top-1 accuracy: The predicted label is the one with the highest probability within the final probability vector. If the class with the highest probability in the prediction is correct, then the prediction is accurate. Otherwise, it is incorrect. This index indicates the number of images that have been correctly classified divided by the total number of test samples.

b) Floating Point Operations per Second: "FLOPs" stands for floating point operations per second, and it is understood as computational workload. It can be used to measure the complexity of algorithms.

c) t-SNE: The purpose of t-distribution stochastic neighbour embedding (t-SNE) is to reduce high-dimensional features to two dimensions, enabling feature visualization.

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TABLE III
TOP-1 ACCURACY (%) ON NWPU-LT

NWPU-LT	Backbone	Imbalance ratio											
		100:1				50:1				10:1			
		Head	Middle	Tail	ALL	Head	Middle	Tail	ALL	Head	Middle	Tail	ALL
Baseline	AlexNet [47]	94.33	84.50	45.35	62.53	94.00	83.67	52.43	68.62	93.33	79.45	68.63	79.37
	VGG-16 [42]	92.00	88.25	57.96	69.27	96.00	88.61	61.61	75.47	92.67	84.10	76.36	83.87
	GoogleNet [43]	97.00	92.00	64.42	76.40	97.75	91.67	70.04	81.16	96.17	90.87	87.38	90.29
	MobileNet-V2 [48]	98.67	93.00	68.08	78.98	98.00	94.83	75.87	85.42	96.50	91.19	84.88	90.78
	ResNet-18 [41]	96.67	92.75	64.58	76.73	98.75	93.17	71.74	82.71	95.50	89.74	82.63	89.24
	ResNet-50 [41]	98.00	94.56	68.77	79.88	98.50	94.72	76.74	85.84	97.00	92.00	84.50	91.33
LDAM [49]	Resnet-50 [41]	98.33	93.50	72.50	81.69	**99.00**	94.06	79.22	86.91	94.33	92.52	87.00	91.78
BKD [50]		98.00	94.88	69.31	80.31	96.50	93.61	81.52	87.69	96.17	91.65	89.88	91.93
CB [19]		95.67	90.75	73.88	81.33	96.00	92.83	79.26	86.18	97.00	92.32	87.13	92.02
LogitAdjustment [51]		**99.33**	95.00	71.65	81.80	98.25	95.22	77.52	86.44	97.17	92.77	88.25	92.56
RSG [52]		98.00	92.69	73.42	81.91	98.50	94.83	78.26	86.69	97.50	92.23	88.88	92.33
Hybrid [27]		97.33	**95.44**	73.58	82.93	97.75	94.61	80.84	87.67	96.17	94.03	88.63	93.36
MiSLAS [40]		97.00	93.38	81.16	86.56	98.25	94.17	85.44	90.07	**97.67**	94.39	**93.38**	94.64
DECOR(ours)	AlexNet [47]	89.67	81.88	58.27	68.76	93.25	87.11	62.04	74.84	92.33	82.94	78.25	83.36
	VGG-16 [42]	96.67	91.25	66.54	77.33	97.25	92.11	71.87	82.22	97.00	88.74	85.75	89.31
	GoogleNet [43]	97.33	93.75	76.54	84.04	97.00	94.39	82.04	88.31	96.83	93.77	89.50	93.42
	MobileNet-V2 [48]	96.33	95.25	79.54	86.24	97.25	95.00	84.96	90.07	96.50	94.06	89.63	93.60
	ResNet-18 [41]	94.67	93.75	77.92	84.67	97.50	94.00	82.83	88.60	96.83	93.13	90.13	93.09
	ResNet-50 [41]	97.33	95.19	**85.38**	**89.67**	97.25	**95.94**	**88.39**	**92.20**	97.50	**95.26**	92.13	**95.00**

TABLE IV
TOP-1 ACCURACY (%) ON AID-LT

AID-LT	Backbone	Imbalance ratio											
		100:1				50:1				10:1			
		Head	Middle	Tail	ALL	Head	Middle	Tail	ALL	Head	Middle	Tail	ALL
Baseline	AlexNet [47]	84.50	73.70	24.72	45.03	85.33	70.45	31.13	50.97	77.50	68.00	47.17	65.10
	VGG-16 [42]	84.50	81.60	28.17	49.73	88.33	75.55	34.56	54.97	83.75	74.10	54.50	71.47
	GoogleNet [43]	92.50	93.20	35.17	58.33	95.67	87.82	40.50	63.37	93.25	83.60	69.00	81.96
	MobileNet-V2 [48]	86.00	90.20	43.56	61.93	94.00	89.18	42.13	64.57	83.60	83.70	71.67	82.57
	ResNet-18 [41]	90.50	91.50	35.78	58.00	91.67	87.36	40.13	62.60	90.25	82.50	66.50	80.33
	ResNet-50 [41]	95.00	91.30	38.94	60.13	95.67	90.36	41.13	64.63	94.00	84.85	72.00	83.50
LDAM [49]	Resnet-50 [41]	87.50	88.60	43.06	61.20	95.33	88.19	54.50	71.13	94.50	84.55	74.33	83.83
LogitAdjustment [51]		91.00	93.90	40.00	61.37	94.33	92.64	47.91	68.90	93.00	85.55	75.67	84.57
CB [19]		90.50	92.70	44.94	63.90	94.33	91.73	46.38	67.80	93.25	86.65	75.67	85.33
MiSLAS [40]		91.50	88.60	50.44	65.90	90.33	86.73	62.75	74.30	87.75	85.30	82.67	85.10
BKD [50]		96.00	**96.20**	51.22	69.20	96.33	91.27	55.19	72.53	93.50	87.85	83.33	87.70
RSG [52]		84.50	94.90	53.94	69.63	91.33	89.27	63.63	75.80	93.75	86.20	77.33	85.43
Hybrid [27]		**96.50**	95.70	52.67	69.93	97.00	94.18	62.56	77.60	**97.75**	87.80	79.83	87.53
DECOR(ours)	AlexNet [47]	81.50	76.20	34.50	51.53	84.00	80.18	39.44	58.83	87.25	75.80	57.83	73.43
	VGG-16 [42]	83.00	83.80	39.77	57.33	88.67	79.36	42.69	60.73	87.25	78.30	62.33	76.30
	GoogleNet [43]	91.00	95.20	68.83	79.10	94.00	94.09	74.38	83.56	95.00	90.75	83.83	89.93
	MobileNet-V2 [48]	94.00	94.60	65.88	77.33	96.33	95.45	75.06	84.67	95.75	90.75	83.67	90.00
	ResNet-18 [41]	91.00	94.20	65.94	77.03	94.00	93.18	75.68	83.93	96.00	89.25	85.33	89.36
	ResNet-50 [41]	95.50	95.80	**72.16**	**81.60**	**97.33**	**95.55**	**79.63**	**87.23**	97.25	**91.75**	**86.83**	**91.50**

TABLE V
TOP-1 ACCURACY (%) ON LONG-TAILED BIT-AFGR50-LT

BOT-AFGR-LT	Backbone	Head	Middle	Tail	ALL
Baseline	AlexNet [47]	97.75	80.10	70.02	72.14
	VGG-16 [42]	98.50	81.10	67.85	70.17
	MobileNet-V2 [48]	99.25	86.5	79.09	80.64
	GoogleNet [43]	99.25	86.7	79.55	81.05
	ResNet-18 [41]	98.75	84.40	77.66	79.18
	ResNet-50 [41]	**99.50**	87.70	**81.09**	82.49
	VIT-B/16 [53]	**99.50**	**97.60**	80.83	**83.25**
DECOR(ours)	AlexNet [47]	99.00	86.60	86.99	87.43
	VGG-16 [42]	98.50	86.00	87.81	88.06
	MobileNet-V2 [48]	99.00	97.20	94.93	94.47
	GoogleNet [43]	**99.50**	97.80	95.13	95.57
	ResNet-18 [41]	99.00	93.70	93.78	93.98
	ResNet-50 [41]	99.25	97.70	**95.78**	**96.11**
	VIT-B/16 [53]	98.50	**98.50**	94.50	95.40

Points with similar features will cluster together, while points with dissimilar features will scatter apart.

d) Confusion matrix: The confusion matrix, in which each column represents a predicted category, and each row represents the true category of the data. The total count in each column represents the number of data predicted as that category, and the total count in each row represents the number of data instances in that category. Each cell represents that the class in its row is predicted as the class in its column.

e) Attention maps: The attention map visualizes the focal points of the model during image processing, in accordance with the Grad-CAM [55] methodology. In the attention map, red regions signify positions with significant features, with deeper shades of red indicating a more substantial contribution to image classification. Blue regions, on the other hand, denote areas lacking crucial features.

C. Comparison Results

We compared the top-1 accuracy of our model with baselines and other long-tail recognition methods, including label-distribution-aware margin loss (LDAM) [49], balanced knowledge distillation (BKD) [50], class-balanced (CB) [19], logit adjustment [51], rare-class sample generator (RSG) [52], hybrid [27], and MiSLAS [40]. The experimental analysis demonstrated that our algorithm outperforms the baseline and

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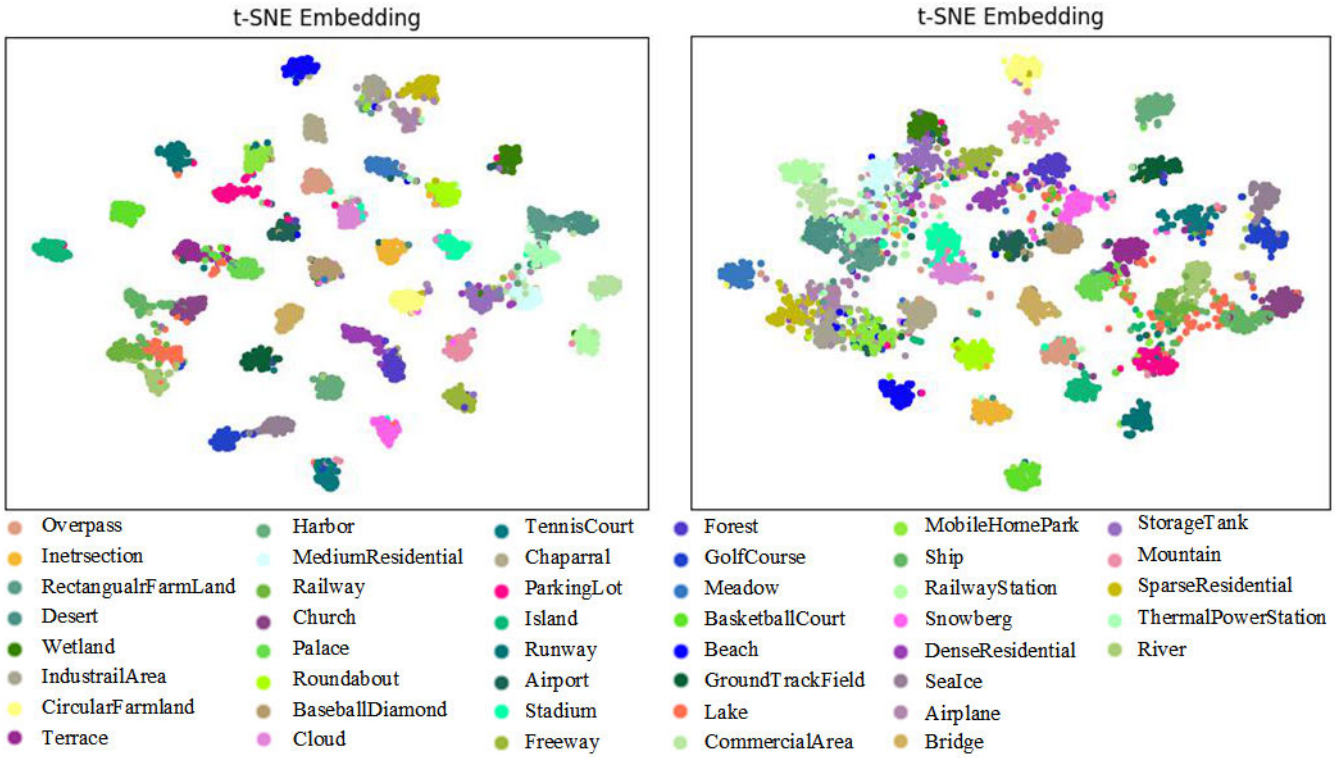


Fig. 6. t-SNE for NWPU-LT with an imbalanced ratio of 0.01 on DECOR (left) and baseline (right).

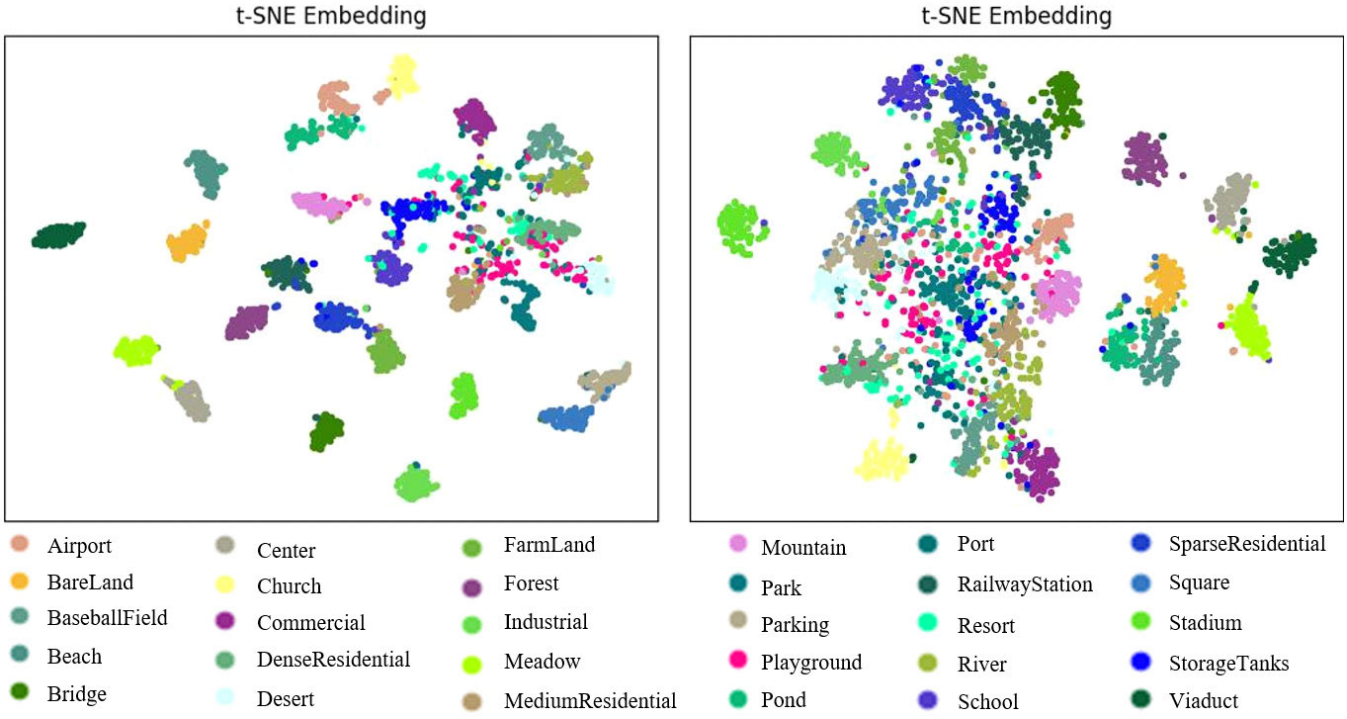


Fig. 7. t-SNE for AID-LT with an imbalanced ratio of 0.01 on DECOR (left) and baseline (right).

all other methods under three different imbalance conditions on two datasets. As shown in Table III, on NWPU-LT, our DECOR algorithm based on ResNet-50 [41] achieved respective improvements of 9.79%, 6.36%, and 3.67% under the 0.01, 0.02, and 0.1 imbalance rates, relative to the baseline based on ResNet-50. This indicates that our algorithm exhibits greater improvements with increasingly imbalanced data. Moreover, our algorithm was also tested using different backbones such as AlexNet [47], VGG-16 [42], GoogleNet [43], MobileNet-V2 [48], and ResNet-18 [41], and all these experiments yielded significant improvements. This suggests that our algorithm's ability to enhance feature representation is universally applicable within the field of neural networks. Furthermore, we compared our method to some existing

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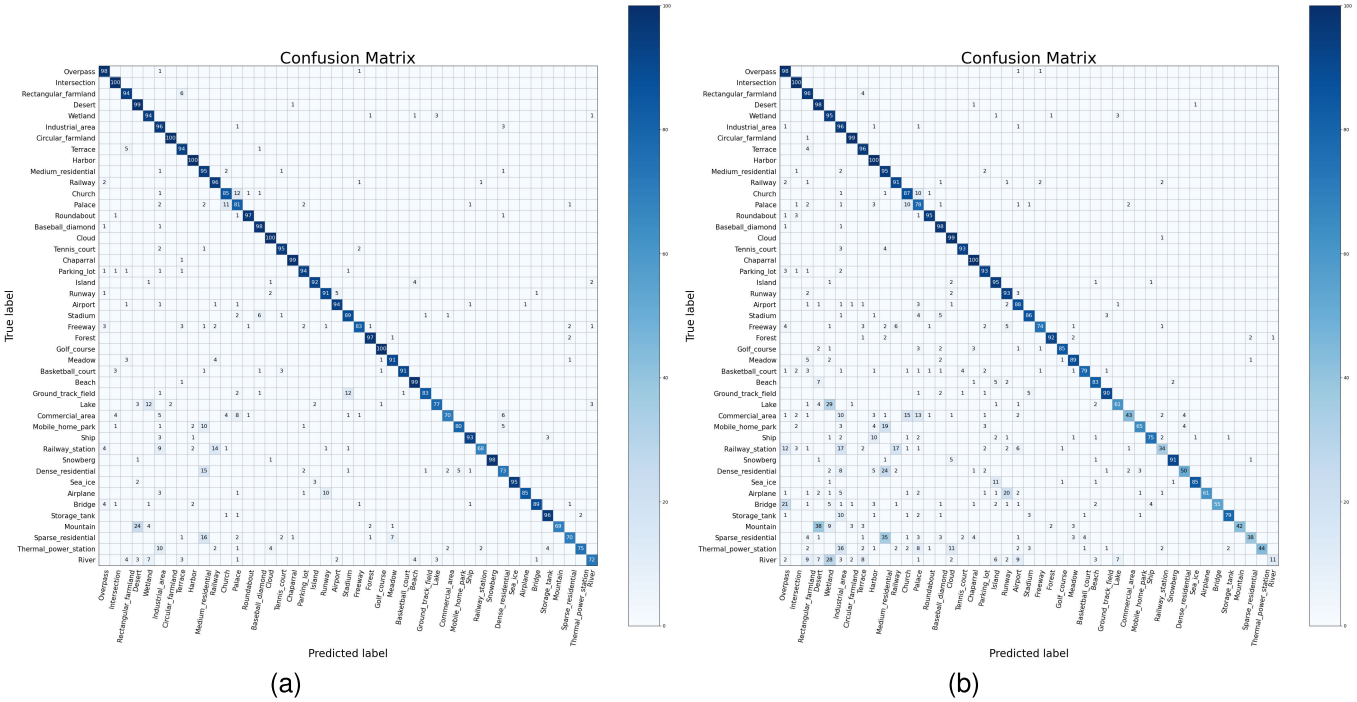


Fig. 8. Confusion matrix for NWPU-LT with an imbalanced ratio of 0.01 on (a) DECOR and (b) baseline.

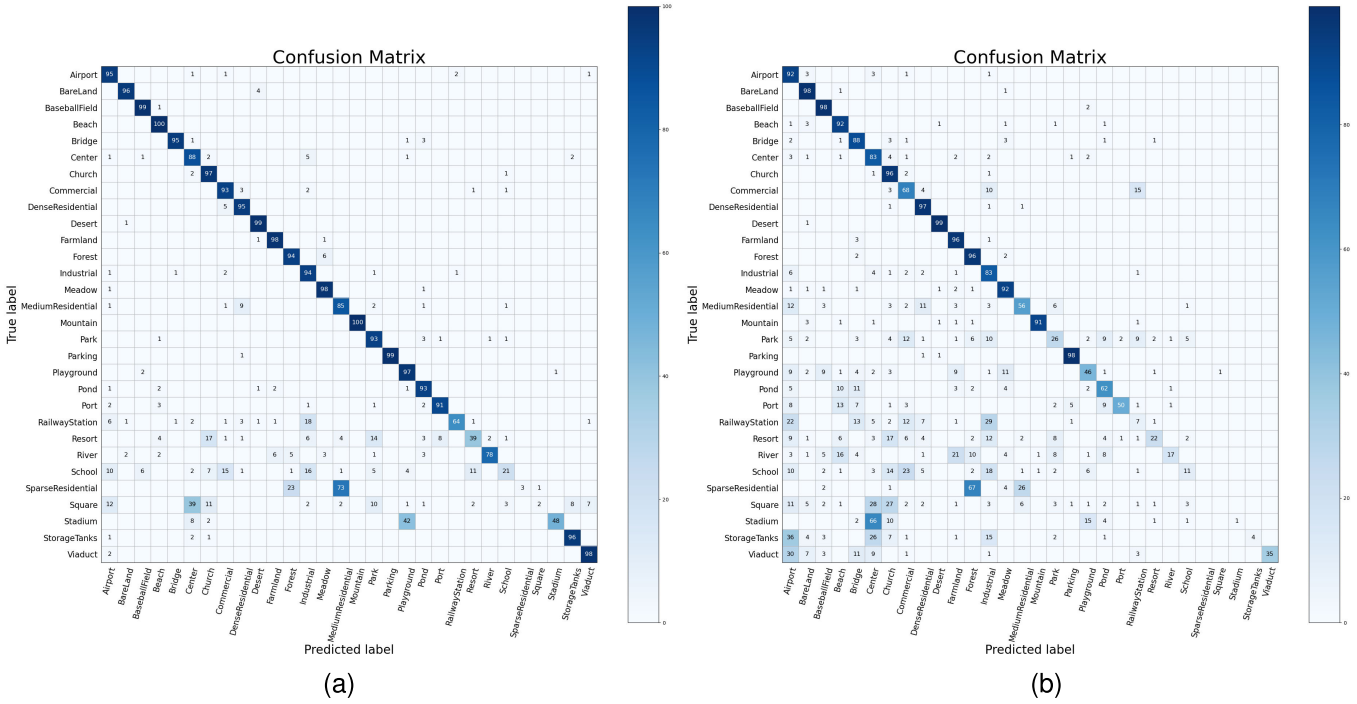


Fig. 9. Confusion matrix for AID-LT with an imbalanced ratio of 0.01 on (a) DECOR and (b) baseline.

long-tail recognition methods, all using ResNet-50 as the backbone for a fair comparison. The results showed that LDAM and CB's loss function methods increased the penalty on tail classes but did not enhance the feature extractor's representation capabilities, resulting in lower accuracy. Logit adjustment provided a method to adjust logit values to focus more on tail classes, while BKD aimed to enhance tail class learning through distillation, with limited improvements in tail accuracy. These methods did not fundamentally address the issue of information scarcity, especially within the tail classes. Although the hybrid method improved the model's representation capabilities, it did not fully resolve the problem of insufficient knowledge in tail classes. The RSG method employs data augmentation on head classes to enhance tail class knowledge, but its ability to enhance knowledge in tail classes is limited. In contrast, our approach not only improves the feature extractor's representation capabilities but also refines knowledge at the data level. The MiSLAS method adopted a completely decoupled two-stage approach to tackle long-tail problems. Our algorithm outperforms it

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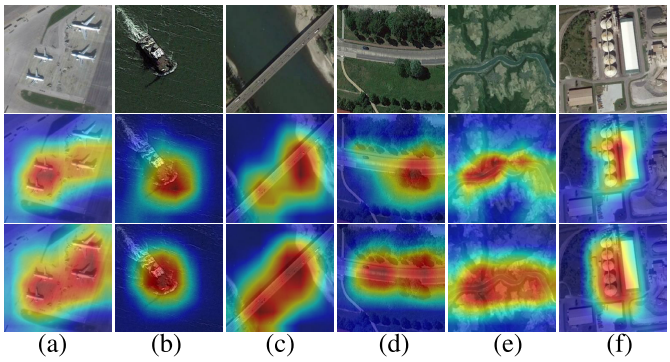


Fig. 10. Visualization of attention maps of baseline (middle) and our DECOR model (bottom). (a) Airplane. (b) Ship. (c) Bridge. (d) Overpass. (e) River. (f) Storage tank.

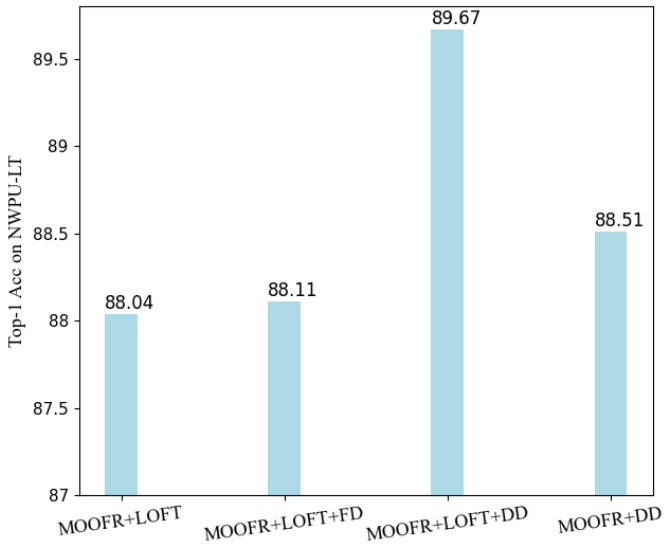


Fig. 11. Top-1 accuracy (%) with different decoupled strategies and the impact of adapter on NWPU-LT with an imbalanced ratio of 0.01.

because our DD method enhances compatibility between the feature extractor and classifier, optimizing the overall performance of the model. As shown in Table IV, our algorithm exhibited similar performance on the AID-LT dataset. Relative to the baseline based on ResNet-50, our DECOR algorithm based on ResNet-50 achieved respective improvements of 21.47%, 22.6%, and 8% under the 0.01, 0.02, and 0.1 imbalance rates. We observed that the accuracy improvement at an imbalance rate of 0.02 was slightly higher than that at an imbalance rate of 0.01. This does not contradict our previous conclusion that "our algorithm exhibits greater improvements with increasingly imbalanced data." This conclusion is predicated on the assumption that there are at least a few samples available for learning in the tail classes. In the case of an imbalance rate of 0.01, many tail classes in the dataset had only one image, indicating extremely limited knowledge, while this situation improved at an imbalance rate of 0.02.

To further validate the effectiveness of our method, we conducted a series of experiments on the more difficult fine-grained long-tailed RSIC dataset, BIT-AFGR50-LT. As shown in Table V, our model demonstrates exceptionally outstanding performance on our self-constructed BIT-AFGR50-LT dataset, achieving an impressive overall top-1 accuracy of 96.11%. In contrast, the baseline model based on ResNet50 achieved an accuracy of 82.49%. Compared to the baseline, our model has shown an improvement of 13.62%. Notably, the middle classes have seen a 10% improvement, while the tail classes have experienced a remarkable 14.69% enhancement. To validate the effectiveness of DECOR on vision transformer, we conducted a set of comparative experiments on the BIT-AFGR50-LT dataset, using VIT-B/16 [53] as the baseline. As shown in the shaded area of Table I, our algorithm achieved an improvement of 12.15%, with a particularly notable increase of 14.06% in the tail classes.

To showcase the outstanding capabilities of the DECOR algorithm, experiments were conducted on diverse datasets. NWPU-LT and AID-LT focused on scene classification, while BIT-AFGR50-LT represented a fine-grained dataset specifically designed for aircraft. Our algorithm exhibited exceptionally impressive performance on both of these distinct dataset types, providing visual evidence of DECOR's ability to generalize across different data categories. However, we posit that there might be potential performance limitations for DECOR in zero-shot tasks. This discovery of potential limitations will be further explored in future research.

To demonstrate that the inference process of our algorithm is extremely straightforward, with few parameters and low model complexity, and does not involve a MOOF, we calculated the FLOPs for baseline, DECOR, and other long-tail recognition algorithms. As shown in Fig. 5, our method can greatly enhance the model's performance even when the FLOPs are nearly identical to those of these six baselines. Furthermore, when compared to other algorithms with similar FLOPs, our model achieves the best performance while maintaining roughly the same level of computational load.

To gain a deeper understanding of the impact of the DECOR method, we visualized the representations learned using baseline cross-entropy and the DECOR training method on NWPU-LT and AID-LT datasets with an imbalance ratio of 0.01. As shown in Figs. 6 and 7, models trained with DECOR typically generate more compact features within the same category, resulting in a more pronounced separation between different categories and improved performance. To visually demonstrate the advantages of the DECOR algorithm, we have provided confusion matrices obtained using both the baseline ResNet-50 and the ResNet-50-based DECOR training method on the NWPU-LT and AID-LT datasets, with an imbalance rate of 0.01. As shown in Fig. 8, for the NWPU-LT dataset, the baseline can only recognize 11 images in the tail class "River," whereas our method can identify 72 images. For the AID-LT dataset in Fig. 9, the baseline can only recognize 1 image in the tail class "Stadium" and four images in the tail class "Storage Tanks." In contrast, our model can recognize 48 and 96 images, respectively. Even in the most challenging classes like "Sparse Residential" and "Square," where the baseline cannot recognize any images, our algorithm successfully identifies images from these two categories, achieving significant breakthroughs. In summary, the DECOR algorithm's overall model performance has seen a significant improvement, with particularly remarkable performance in tail class recognition.

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TABLE VI
COMPONENTWISE ABLATION EXPERIMENTAL RESULTS (%) ON NWPU-LT WITH IMBALANCED RATIO OF 0.01

Method	Module			Head	Middle	Tail	ALL
	CB-RS (M1)	LFC (M2)	WKL (M3)				
Baseline				98.00	94.56	68.77	79.88
M1	✓			98.00	94.69	68.27	79.64(-0.24)
M2		✓		97.67	94.69	74.15	83.02(+3.14)
M1+M2	✓	✓		96.33	95.31	79.65	86.33(+6.45)
M2+M3		✓	✓	97.67	95.44	77.35	85.13(+5.25)
M1+M2+M3	✓	✓	✓	96.33	95.63	80.96	87.20(+7.22)

To demonstrate the robust representational capabilities of our model, we visualized attention maps generated by both the baseline and the DECOR models. In comparison to the baseline, our model's attention maps exhibit a more accurate and comprehensive focus on crucial regions of the images. As depicted in Fig. 10(a), our model simultaneously attends to all four airplanes, whereas the baseline only captures attention on three. Illustrated in Fig. 10(b), the DECOR model attends to the entire ship, whereas the baseline focuses solely on the front half of the ship. Consequently, the feature maps showcase the enhanced representational capabilities of our algorithm in image processing.

The design of the DECOR algorithm prioritizes generality and transferability. Utilizing a modular structure enables it to seamlessly adapt to changes in model architecture when confronted with different tasks. This flexibility allows the algorithm to not only be confined to classification tasks but also to effortlessly extend into areas such as object detection and segmentation. Serving as a plug-and-play module, the universality of this architecture facilitates the model to easily play a role in handling various perceptual tasks. Consequently, we plan to introduce the DECOR model into further research in the domains of object detection and segmentation. However, concerning zero-shot capabilities, we acknowledge some limitations in our algorithm's ability to recognize unknown objects, which will be a focal point of our future investigations.

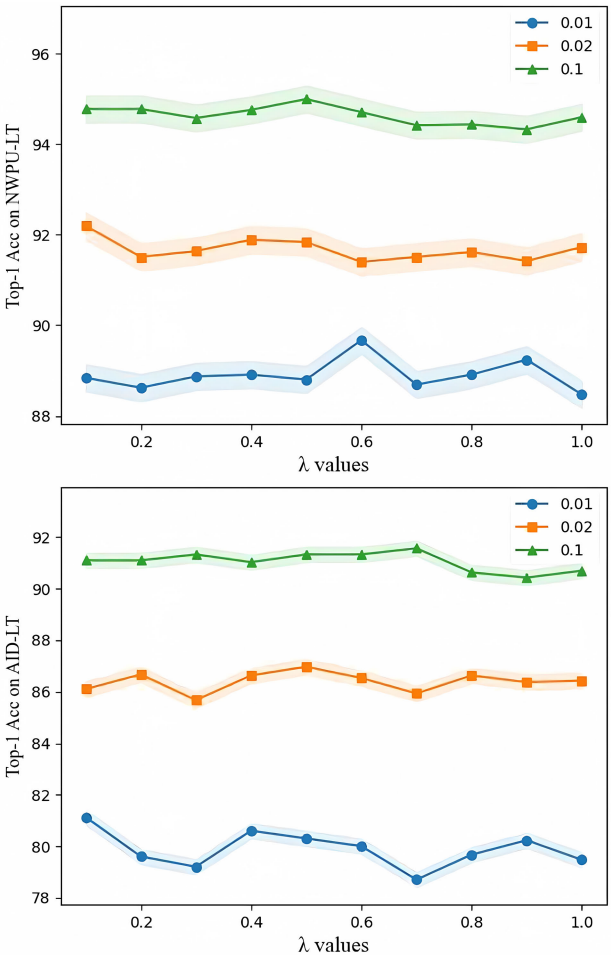


Fig. 12. Top-1 accuracy (%) with different values of residual factor λ of DECOR on NWPU-LT (top) and AID-LT (bottom) with an imbalanced ratio of 0.01 (blue), 0.02 (orange), 0.1 (green).

D. Ablation Studies

In this section, we conduct ablation studies to evaluate the individual contributions of various components in our proposed DECOR. The analysis aims to provide a deeper understanding of the model's performance and identify areas for potential improvement.

1) Componentwise Analysis: First, we begin by analyzing the impact of each component on the model's overall performance. As shown in Table VI, we select ResNet-50 [41] as the baseline recognition method and *class balancing resampling* (CB-RS) denotes the baseline method with class balancing resampling strategy. *LFC* refers to our proposed LFC, and *WKL* represents the WKL. In addition, as shown in Fig. 11, MOOFR represents the proposed MOOF with class-balanced resampling, and *LOFT* stands for lightweight optimization fine-tuning technique. *FD* denotes fully decoupled framework, and *DD* represents the proposed dynamic decoupling strategy.

The experiments are structured as follows: starting with the ResNet-50 baseline, we incrementally integrate each component. The performance metrics are recorded for head, middle, tail, and ALL categories, following the long-tail distribution pattern. The detailed componentwise results are presented in Table VI and Fig. 11.

a) Baseline performance: The baseline model with ResNet-50 achieved a performance of 98.00% in the head category, 94.56% in the middle category, 68.77% in the tail category, and an overall accuracy of 79.88%. This set the stage for subsequent improvements through the integration of DECOR components.

b) Impact of CB-RS: Integrating CB-RS with the baseline resulted in the middle category's accuracy slightly increasing to 94.69% and the tail category's accuracy marginally decreasing to 68.27%, with a slight decrease in

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overall performance to 79.64% (−0.24%). This highlights the importance of addressing the class imbalance in RSIC, especially in long-tail distributions. However, due to the principle that better features result in a better classifier, solely relying on CB-RS may not effectively address long-tail classification problems.

c) Addition of LFC: The inclusion of LFC led to a substantial improvement in the tail category, with the accuracy increasing to 74.15%. The head category saw a slight decrease in performance to 97.67%, and the middle category remained stable at 94.69%. The overall accuracy improved to 83.02% (+3.14%), indicating the effectiveness of LFC in enhancing feature representation for under-represented classes.

d) Synergistic effect of CB-RS and LFC: When CB-RS and LFC were combined, the head category's performance decreased to 96.33%, while the middle and tail categories improved to 95.31% and 79.65%, respectively. The overall performance saw a significant boost to 86.33% (+6.45%), underscoring the complementary nature of these techniques in tackling the challenges of long-tail distribution. Since CB-RS is used in conjunction with LFC, not only the learning of the classifiers can be more balanced, but also the feature representation ability of the network can be significantly improved. Therefore, there is a significant performance leap when CB-RS and LFC are used jointly compared to CB-RS alone.

e) Integration of WKL: The addition of WKL, combined with LFC, referred to as MOOF, further enhanced performance, especially in the middle and tail categories, which increased to 95.44% and 77.35%, respectively. The overall performance was also improved to 85.13% (+5.25%), demonstrating the significant role of external knowledge in augmenting the model's generalization capabilities across different categories.

f) Collaborative influence of CB-RS, LFC, and WKL: When jointly combining CB-RS, LFC, and WKL (referred to as MOOFR), the performance of the head class remains unchanged compared to the combination of only CB-RS and LFC, while the middle and tail classes exhibited improvements of 0.32% and 1.31%, respectively. The overall performance saw an enhancement of 0.77%. This reaffirms the reinforcing effect of WKL on tail classes with limited samples. In comparison to the combination of only LFC and WKL, the performance of the head class decreased by 1.34%, while the middle and tail classes experienced respective improvements of 0.19% and 3.61%. The overall performance demonstrated a significant enhancement of 2.07%. These findings underscore the crucial impact of the joint integration of CB-RS, LFC, and WKL on the model's performance.

g) Decoupling approaches and LOFT: As depicted in Fig. 11, we compare three training approaches: nondecoupling, FD, and DD. DD ensures compatibility between the feature extractor and the classifier, resulting in a significant performance improvement to 89.67%. The LOFT also significantly contributes to the model's performance. In comparison to the performance without the inclusion of the LOFT module (88.51%), the model exhibited a remarkable improvement of 1.16%.

h) Conclusive insights: These results clearly demonstrate the individual and collective strengths of the DECOR components in improving long-tail classification performance. Each component, namely, CB-RS, LFC, WKL, DD, and LOFT not only addresses specific challenges in RSIC but also synergistically enhances the overall model performance, especially in the under-represented tail category. The DECOR framework, therefore, stands as a robust approach to tackling the inherent challenges of long-tail distribution in RSIC.

E. Effectiveness of LOFT

Our empirical investigations into the LOFT technique reveal its efficacy in enhancing the model's performance across various settings. Fig. 12 illustrates the top-1 accuracy with different values of the residual factor λ for DECOR on NWPU-LT and AID-LT. These figures provide a clear visualization of how LOFT contributes to accuracy improvement when the residual factor is tuned optimally.

As evidenced by the peak accuracies observed at λ values of 0.6 for NWPU-LT [Fig. 12(top)] and 0.8 for AID-LT [Fig. 12(bottom)], LOFT demonstrates a pronounced impact on the model's classification capabilities. The optimal value of λ signifies the balance between the original features and the additional lightweight parameters introduced during fine-tuning. It is noteworthy that the accuracy increment is not linear with the increase of λ, suggesting a complex relationship between feature adaptation and the learning process.

For NWPU-LT, the top-1 accuracy incrementally increases with λ, reaching a peak at 0.6 before a gradual decline, indicating an optimal tradeoff at this value where the additional parameters contribute most effectively to the model without overwhelming the original features. Conversely, the AID-LT dataset shows a more direct correlation between λ and accuracy up to the 0.8 mark, beyond which the performance slightly tapers off.

These observations from both datasets underscore the critical role of LOFT in adapting the model to the specificities of the data distribution. By carefully calibrating the residual factor, LOFT allows for a refined balance between preserving the integrity of pretrained features and introducing new adaptations, ultimately leading to a tangible improvement in classification accuracy. The results firmly establish LOFT as an indispensable part of our DECOR framework, particularly beneficial in scenarios where precision fine-tuning is essential to address the nuances of long-tail distribution in RSIC.

V. Conclusion

In this article, we introduce a DD framework, redirecting the focus of our model training from the smooth transition of representation learning to classifier learning. In addition, the parameters of the feature extractor undergo dynamic fine-tuning through a multiobjective optimization strategy to enhance feature representation. Furthermore, we employ class-balanced resampling techniques in conjunction with parameter-efficient adapters, balancing the classification decision boundary and improving the alignment between the feature extractor and the classifier. DECOR has proven to be

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an outstanding solution for addressing the challenges of long-tailed distributions, encompassing issues of class imbalance and few shots. These challenges are pervasive in real-world remote sensing images, making the success of DECOR significant for open-world RSIC. However, we acknowledge that our algorithm exhibits suboptimal performance in identifying unknown categories in zero-shot tasks, which will be a key focus for future research. Furthermore, as a plug-and-play module, we plan to extend DECOR to remote sensing visual tasks such as object detection and segmentation. This series of findings provides valuable directions for further research in this field.

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